

AHMAD FAIQ FAWWAZ

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SUMMARY

Computer Engineering student at Sepuluh Nopember Institute of Technology (ITS) with 2+ years of hands-on experience in **Robotics and Embedded Systems**. As **Senior Programmer for VI-ROSE ITS**, I architect robotic control systems and motion-planning applications, optimizing firmware to achieve peak performance in national competitions. Proficient in **C++, C, Python, and RTOS**, with a strong foundation in digital circuit design, IoT ecosystem development, and technical mentoring.

EXPERIENCE

VI-ROSE ITS Robotics Team | *Programming Staff* | Feb 2024 – Present

- Developed and designed **robotic control systems** for competition, ensuring robust **hardware-software integration**, leading the team to secure **1st Place at National and Regional levels (KRSTI 2024)**.
- Engineered a custom App for robot motion, focusing on **user-centric technical solutions** and movement precision.
- Played a key role in optimizing system performance through **efficient troubleshooting** and precise programming implementation.

B401 Robotics Laboratory | *Laboratory Assistant & Practicum Mentor* | Mar 2025 – Present

- Laboratory Assistant (Aug 2025 – Present):** Manage laboratory resources and facilitate research in robotics and embedded systems.
- Digital Circuit Practicum Assistant (Mar 2025 – Jun 2025, Aug 2025 – Dec 2025, Mar 2026 – Present):** Mentored 100+ students in digital logic, Boolean optimization, and sequential circuit design using 74LS series ICs.
- Telematics Workshop Assistant (Aug 2025 – Nov 2025, Mar 2026 – Present):** Led workshops on **soldering, PCB design, 3D printing**, and basic embedded firmware programming.
- Evaluated student performance and provided constructive feedback to strengthen logical problem-solving skills.

LAB MIOT (Multimedia and IoT) | *Programming Practicum Assistant* | Sep 2024 – Dec 2024

- Mentored students in foundational programming concepts (C).
 - Assisted in designing lab exercises to improve students' algorithmic thinking and logical problem-solving.
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EDUCATION

Institut Teknologi Sepuluh Nopember (ITS) | 2023 – 2027 (Expected)

- Undergraduate Computer Engineering Student | **GPA:** 3.65/4.00

SMAN 2 Sidoarjo | 2020 – 2023

TECHNICAL PROJECTS

Smart Power System Monitoring with ML | *ESP32, PZEM-004, Node-RED, Prometheus, Grafana* | Sep – Dec 2025

- Developed an IoT-based power monitoring system using ESP32 and PZEM-004, utilizing **MQTT/HTTP protocols** for high-reliability data acquisition.
- **Architected the data pipeline using Node-RED** as a centralized orchestrator to bridge real-time sensor data with **Prometheus** (time-series database) and **Grafana**.
- Integrated a **Machine Learning model** for device classification, enabling automated identification of electrical loads within an industrial monitoring dashboard.

Computer Vision Card Game (AR Math-Puzzle) | *Python, OpenCV, Raylib, C++* | Sep – Dec 2025

- Built a CV pipeline using **HSV masking, perspective warping, and template matching**.
- Synchronized a **multi-threaded C++ engine** to process real-time physical card data as mathematical operators.

LED Matrix Handheld Game | *ESP-IDF, FreeRTOS, ESP32* | Sep – Dec 2025

- Implemented a **dual-core task distribution** strategy for zero-latency gameplay.
- Leveraged low-level **GPIO register manipulation** and NVS (Non-Volatile Storage) for high-score tracking.

Home Assistant | *Lead Embedded & AI Developer* | Aug – Dec 2025

- **Programmed and integrated Raspberry Pi and ESP32** as the core controller and edge nodes, establishing a seamless local-cloud communication ecosystem via Wi-Fi.
- **Implemented WISPR AI** for voice-command processing, allowing users to control home appliances through high-accuracy natural language transcription.
- **Developed a Machine Learning model** to analyze power consumption data, providing predictive analytics to **calculate and estimate monthly electricity bills** based on real-time usage.

Visual Servoing on Quadraped Robot | *C++, OpenCV, Robotics Control* | Nov – Dec 2025

- Implemented **Image-Based Visual Servoing (IBVS)** for autonomous adjustment of robot position.
- Achieved perfect alignment and distance for robust autonomous grasping.

Wall-Avoider Mobile Robot | *ESP32, Ultrasonic Sensors, L298N* | Aug – Nov 2025

- Developed an autonomous robot using five ultrasonic sensors for precise real-time obstacle avoidance.

- Engineered a **differential drive locomotion model** for seamless movement in confined spaces.

Data Structure & Algorithm Boss Fight Game | C++, Raylib | **Oct – Dec 2024**

- Implemented an **Object Pooling Pattern** to eliminate runtime memory allocation overhead.
- Optimized CPU cache utilization through contiguous memory structures for complex AI behaviors.

Hardware-based Finite State Machine (FSM) | 74LS ICs, JK Flip-flops, NE555 | **Oct – Dec 2024**

- Designed a synchronized two-digit 7-segment display sequence (17-37-57-77).
- Utilized **Boolean optimization** for state transitions and an NE555 timer for adjustable speed.

IoT Distance Target Game | Embedded Systems Developer | **Jun – Jul 2024**

- **Engineered the embedded hardware layer** utilizing ESP32 to handle real-time distance tracking and game logic processing at the edge.
- Developed firmware to interface ultrasonic sensors with a Node.js/Express backend, ensuring low-latency data transmission for a responsive gaming experience.

Automatic Sorting Robot ARM | Robotics Programmer | **Feb – Mar 2024**

- Programmed an **autonomous sorting system** using robot arm, ultrasonic and TCS color sensors for real-time object classification.

SKILLS

- **IoT Protocols:** MQTT, HTTPs REST API, Serial UART, I2C, SPI.
- **IDEs & Tools:** Node-RED, Visual Studio Code, PlatformIO, Git/GitHub, Docker.
- **Microcontrollers:** ESP32 (ESP-IDF/Arduino), Raspberry Pi, STM32, Arduino.
- **Programming:** C++, C, Python, JavaScript, TypeScript, HTML/CSS.
- **Soft Skills:** Technical Support, Troubleshooting, English & Bahasa Communication, Independent & Team Work.

HONORS & AWARDS

- **1st Place** – National Indonesian Dance Robot Contest (KRSTI) | **2024**
- **1st Place** – Regional Level Indonesian Dance Robot Contest (KRSTI) | **2024**

CERTIFICATIONS

- **Legacy Javascript Algorithms and Data Structures** | FreeCodeCamp | **2024** | www.freecodecamp.org/certification/fawzqi/javascript-algorithms-and-data-structures-v8

freeCodeCamp (f)

This certifies that
Ahmad Faiq Fawwaz
successfully completed the
Legacy JavaScript Algorithms and Data Structures V8
Developer Certification on January 5, 2024
representing approximately 300 hours of work




Quincy Larson
Executive Director, freeCodeCamp.org



Verify this certification at:
<https://freecodecamp.org/certification/fawwaz/javascript-algorithms-and-data-structures-v8>

INSTITUT TEKNOLOGI SEPULUH NOPEMBER
TRANSKRIP SEMENTARA/TEMPORARY ACADEMIC TRANSCRIPT



FAKULTAS TEKNOLOGI ELEKTRO DAN
INFORMATIKA CERDAS
FACULTY OF INTELLIGENT ELECTRICAL
AND INFORMATICS TECHNOLOGY

Departemen / : Teknik Komputer / Computer Engineering
Nama / Name : Ahmad Faiq Fawwaz
NRP / ID No : 5024231032
Tempat, Tanggal Lahir / Place, Date of Birth : Sidoarjo, 28 Juli 2005

Indeks Prestasi / GPA : 3,65
Tahun Masuk / Entrance Year : 2023

No	Kode Code	Mata Kuliah Subject	Sem Sem	Kr Gr	Nilai Grade	No	Kode Code	Mata Kuliah Subject	Sem Sem	Kr Gr	Nilai Grade
1	EC234101	Dasar Pemrograman Basic Programming	1	3	A	22	EC234403	Jaringan Komputer Computer Network	4	3	B
2	EC234102	Aljabar Linear Linear Algebra	1	3	A	23	EC234405	Sistem Operasi Operating Systems	4	3	AB
3	EC234103	Probabilitas dan Statistika Probability and Statistics	1	3	AB	24	EC234406	Sistem Mikroprosesor dan Mikrokontroler Systems of Microprocessor and Microcontrollers	4	3	AB
4	EC234101	Pengantar Teknologi Elektro dan Informatika Cerdas Introduction to Intelligent Electrical and Informatics	1	2	AB	25	EC234407	Pengantar Robotika Introduction to Robotics	4	3	AB
5	EC234103	Fisika Mekanika Mechanics Physics	1	3	A	26	EC234501	Pembelajaran Mesin Machine Learning	5	3	A
6	EC234101	Kalkulus 1 Calculus 1	1	3	A	27	EC234502	Keamanan Sistem Komputer Computer System Security	5	3	AB
7	EC234201	Pemrograman Lanjut Advanced Programming	2	3	AB	28	EC234503	Pengolahan Citra dan Video Digital Image and Video Processing	5	3	AB
8	EC234202	Metode Numerik Numerical Methods	2	2	A	29	EC234504	Sistem Terapan Engineering Systems	5	3	AB
9	EC234203	Matematika Diskrit Discrete Mathematics	2	3	B	30	EC234505	Desain dan Aplikasi Internet of Things Design and Application of Internet of Things	5	3	AB
10	EC234204	Desain Dan Rekayasa Sistem Systems Design and Engineering	2	2	A	31	EC234541	Sistem Konsumsi Waktu Nyata Real-Time Consumption Systems	5	3	AB
11	EC234205	Rangkaian Listrik Electric Circuits	2	2	AB	32	EC234542	Robot Berjalan Robotical Robot	5	3	AB
12	EC234202	Fisika Listrik Dan Magnet Physics of Electricity and Magnetism	2	4	AB	33	EC234603	Kompetensi Teknik Komputer 1 Computer Engineering Competencies 1	6	2	A
13	EC234201	Kalkulus 2 Calculus 2	2	3	AB	34	UG234932	Kebudayaan dan Kebangsaan Culture and Nationality	6	3	AB
14	EC234301	Rangkaian Digital Digital Circuit	3	3	AB						
15	EC234302	Persamaan Differensial dan Deret Differential Equations and Series	3	4	AB						
16	EC234303	Struktur Data dan Analisis Algoritma Data Structures and Algorithm Analysis	3	3	A						
17	EC234304	Pengolahan Sinyal Digital Digital Signal Processing	3	3	B						
18	EC234305	Kalkulus Banyak Variabel Multivariable Calculus	3	4	AB						
19	EC234306	Pengantar Elektronika Electronic Circuit	3	3	A						
20	EC234401	Sistem Manajemen Basis Data Database Management Systems	4	3	B						
21	EC234402	Analisis dan Organisasi Sistem Komputer Computer System Architecture and Organization	4	3	A						
Jumlah Kredit / Total of Credits										100	

- This document is only use for: student exchange, short program, internship program, scholarship, joint/double degree, and registration to master degree.
- Should any data differences occur, then the valid data will refer to Online Academic Information System

Institut Teknologi Sepuluh Nopember

Kode	Grade Explanation (Points)
A	Excellent (4)
AB	Very Good (3.5)
B	Good (3)
BC	Sufficient (2.5)
C	Fair (2)
D	Poor (1)
E	Very Poor (0)

Surabaya, 17 April 2025
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